Varoujan KARAKOZIAN

AI Engineer Apprentice at Dassault Systèmes

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Profile

Second-year engineering student at EFREI Paris, specializing in **Big Data & Machine Learning (BDML)**. Currently on a three-year work-study program (2025–2028) with **Dassault Systèmes**, in an R&D DevSecOps team building AI tooling for internal developers, working with **MCP** and **RAG**.

Education

Engineering Cycle, Big Data & Machine Learning specialization
EFREI Paris (work-study) *2025 – 2028*
2nd year in progress.

International Preparatory Classes (English track)
EFREI Paris

Baccalauréat (French high-school diploma), High Honors
Lycée Sainte-Marie, Créteil

Experience

AI Engineer Apprentice
Dassault Systèmes, Vélizy-Villacoublay, France *Sept. 2025 – Jun 2028*

- Built **MCP** (Model Context Protocol) tools connecting an AI coding assistant to internal systems: automatic version-control context detection, build metadata retrieval, automated web-scraping data collection (Selenium).
- Designed and evaluated a local **RAG** pipeline (BM25) for internal documentation indexing, replacing an LLM-based approach that was 2–3 orders of magnitude slower and more token-expensive.
- Built a three-model evaluation protocol (question generation, answering, LLM-as-judge) measuring retrieval Accuracy@K across roughly 8,500 questions over 3 benchmark corpora.
- Contributed to **FixGen**, a generative-AI crash diagnosis tool: built data-collection tools (URL classification, automated stack-trace scraping).

Sales Internship
Lacoste, Créteil, France *2025 (1 month)*

- Welcomed and advised customers, providing personalized recommendations tailored to their needs.
- Used language skills (English, Spanish) to assist international customers.

Warehouse Internship (Order Picker)
Phoenix Pharma, Créteil, France *2024 (1 month)*

- Logistics: preparing and checking pharmaceutical orders.
- Teamwork: effective collaboration with warehouse teams in a fast-paced environment.

Academic Projects

NYC Taxi: Data Visualization & Business Intelligence *2026*

- Exploratory analysis of NYC taxi trips (Python, Pandas, Plotly, Streamlit): data cleaning, interactive visualizations, BI dashboard with KPIs.

Wine Quality Classification: Machine Learning *2026*

- Compared multiple models (Random Forest, SVM, etc.) to predict wine quality from physicochemical features.

Earlier projects: Climate Change Data Analysis (2025): full PCA pipeline showing a correlation between inequality and national pollution. **Voll'Effrei** (2025): volleyball video game, interdisciplinary physics/CS project. **Intelligent Chatbot** (2024): Python chatbot based on TF-IDF analysis. **CDataFrame** (2024): a C-language DataFrame inspired by Pandas.

Skills

AI / Machine Learning
- MCP (Model Context Protocol)
- RAG & RAG evaluation
- LLM-as-judge, Accuracy@K
- BM25, scikit-learn (Random Forest, SVM)

Programming
- Python, C, Java

Tools & platforms
- Git/GitHub, Linux
- Selenium, Streamlit, Plotly, Pandas

Databases
- SQL, MySQL, SQLite

Soft skills
- Teamwork
- Adaptability
- Self-taught
- Initiative
- Creative
- Autonomous
- Versatile

Languages

- French: Native
- English: Advanced (B2)
- Armenian: Intermediate (B1)
- Spanish: Intermediate (B1)

Hobbies

- Music
- Climbing
- Chess
- Basketball